Fitting logistic regression models

Last time: Class activity

$$\pi = P(Accepted = 1)$$

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -8.712 + 0.246\,\text{MCAT}$$

> What is the change in the odds of acceptance associated with an increase of 1 point on the MCAT?

$e^{0.246} = 1.28$

An increase of 1 point on the MCAT is associated with an increase in odds of acceptance by a factor of 1.28

i.e., a 28% increase in odds of acceptance

Class activity

$$\pi = P(Accepted = 1)$$

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -8.712 + 0.246 \text{ MCAT}$$

> What is the estimated probability that a student with an MCAT score of 40 is accepted?

$$\hat{\pi} = \frac{\text{odds}}{1+\text{odds}} = \frac{\exp\{-8.712 + 0.246(40)\}}{1+\exp\{-8.712+0.246(40)\}} = 0.76$$

i.e. 76% chance of being accepted for a student w/ an MCAT score of 40

Class activity

$$\pi = P(Accepted = 1)$$

$$\log\left(\frac{\widehat{\pi}}{1-\widehat{\pi}}\right) = -8.712 + 0.246\ \mathrm{MCAT}$$

> For approximately what MCAT score would a student have a roughly 50-50 chance of being accepted to medical school?

want MCAT score $\rightarrow$ $\hat{\pi} = 0.5$

$\hat{\pi} = 0.5 \Rightarrow \frac{\hat{\pi}}{1-\hat{\pi}} = 1 \Rightarrow \log\left(\frac{\hat{\pi}}{1-\hat{\pi}}\right) = 0$

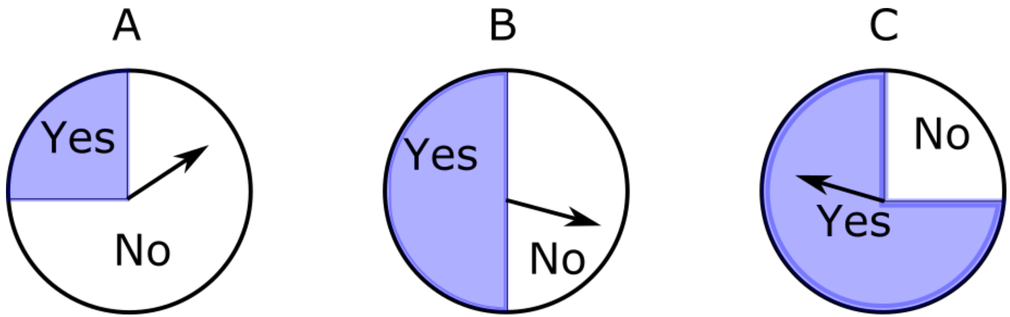
$0 \overset{\text{set}}{=} -8.712 + 0.246\ \text{MCAT}$

$\Rightarrow \text{MCAT} = \frac{8.712}{0.246} = 35.4$

Warmup

Work on the warmup activity (handout).

Warmup

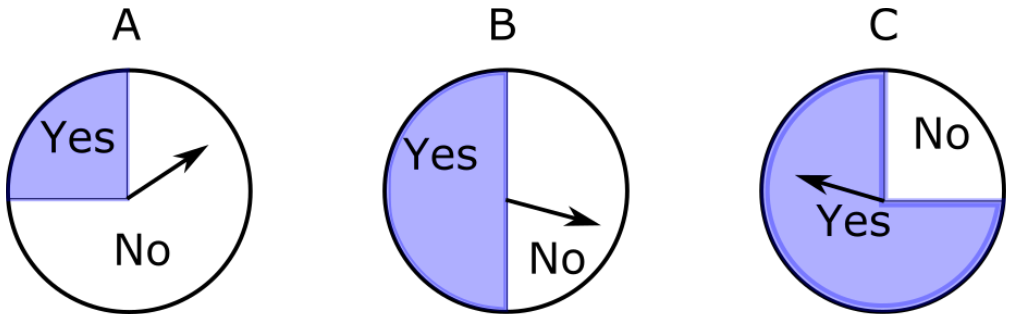


$\pi = P(Yes) = \frac{1}{4}$ $\qquad$ $\pi = \frac{1}{2}$ $\qquad$ $\pi = \frac{3}{4}$

A spinner is chosen and spun twice, resulting in No, No.

- $P(\text{No}, \text{No})$ if $\pi = \frac{1}{4}$: $\quad \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{16}$
- $P(\text{No}, \text{No})$ if $\pi = \frac{1}{2}$: $\quad \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = \frac{4}{16}$
- $P(\text{No}, \text{No})$ if $\pi = \frac{3}{4}$: $\quad \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$

Warmup

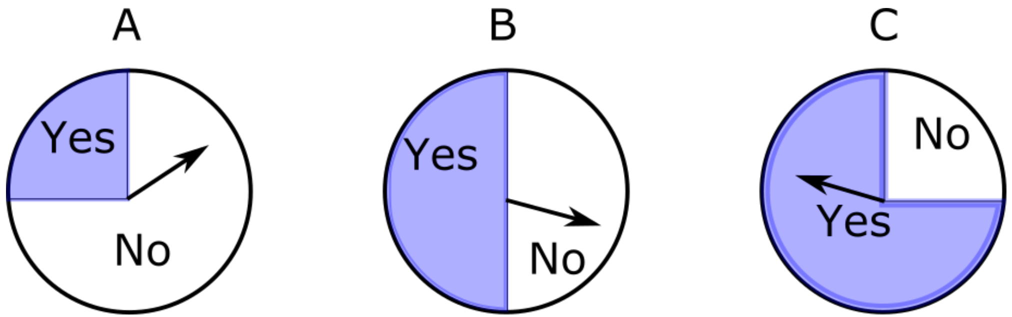


$\pi = P(Yes) = \frac{1}{4}$ $\qquad \pi = \frac{1}{2}$ $\qquad \pi = \frac{3}{4}$

A spinner is chosen and spun twice, resulting in No, No.

- $P(\text{No, No})$ if $\pi = \frac{1}{4}$: $\left(\frac{3}{4}\right)\left(\frac{3}{4}\right) = \frac{9}{16}$
- $P(\text{No, No})$ if $\pi = \frac{1}{2}$:
- $P(\text{No, No})$ if $\pi = \frac{3}{4}$:

Warmup



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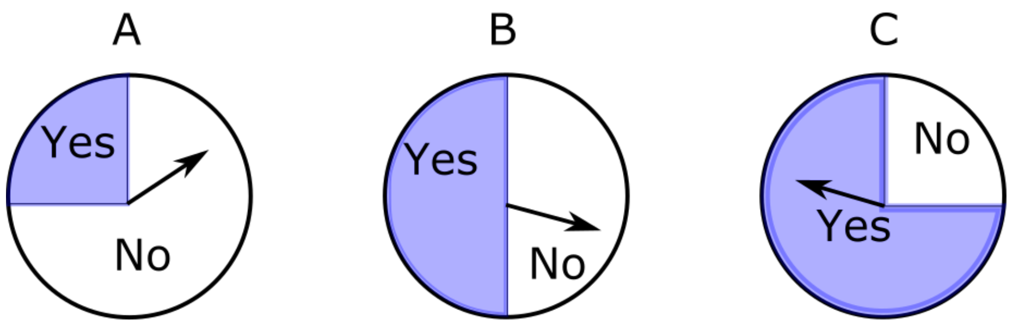
- $P(\text{No, No})$ if $\pi = \frac{1}{2}$: $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{4}{16}$
- $P(\text{No, No})$ if $\pi = \frac{3}{4}$:

Warmup



$\pi = P(Yes) = \frac{1}{4}$ $\qquad \pi = \frac{1}{2}$ $\qquad \pi = \frac{3}{4}$

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Warmup

A spinner is chosen and spun twice, resulting in No, No.

- $P(\text{No, No})$ if $\pi = \frac{1}{4}$: $\left(\frac{3}{4}\right)\left(\frac{3}{4}\right) = \frac{9}{16}$
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- $P(\text{No, No})$ if $\pi = \frac{3}{4}$: $\left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \frac{1}{16}$

Which spinner would you guess was spun?

Warmup

A spinner is chosen and spun twice, resulting in No, No.

- $P(\text{No}, \text{No})$ if $\pi = \frac{1}{4}$: $\left(\frac{3}{4}\right)\left(\frac{3}{4}\right) = \frac{9}{16}$
- $P(\text{No}, \text{No})$ if $\pi = \frac{1}{2}$: $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{4}{16}$
- $P(\text{No}, \text{No})$ if $\pi = \frac{3}{4}$: $\left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \frac{1}{16}$

Which spinner would you guess was spun? **Spinner A, because the probability of the observed data is the highest**

Warmup

A spinner is chosen and spun four times, resulting in No, Yes, Yes, No.

- $P(\text{No}, \text{Yes}, \text{Yes}, \text{No})$ if $\pi = \dfrac{1}{4}$: $\quad \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{9}{256}$
- $P(\text{No}, \text{Yes}, \text{Yes}, \text{No})$ if $\pi = \dfrac{1}{2}$:
- $P(\text{No}, \text{Yes}, \text{Yes}, \text{No})$ if $\pi = \dfrac{3}{4}$:

Warmup

A spinner is chosen and spun four times, resulting in No, Yes, Yes, No.

- $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{1}{4}$: $\frac{9}{256}$
- $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{1}{2}$: $\frac{16}{256}$
- $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{3}{4}$: $\frac{9}{256}$

Which spinner would you guess was spun?

Warmup

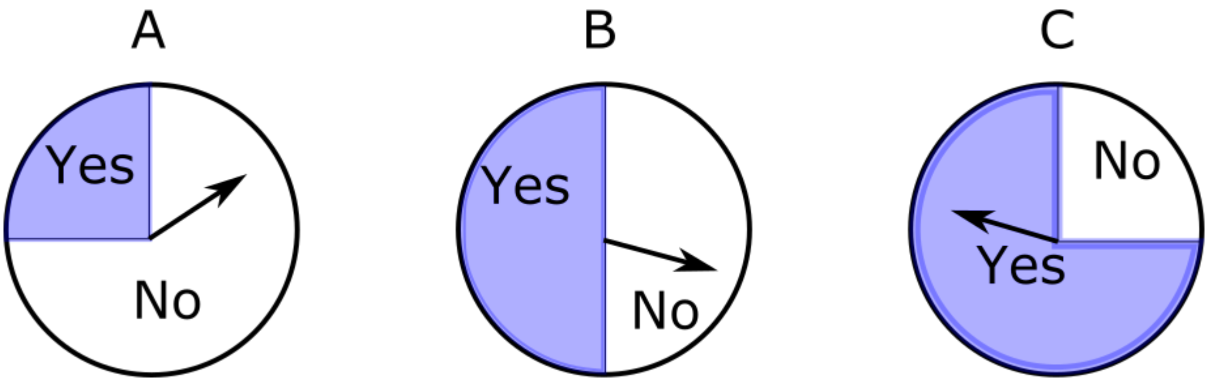
A spinner is chosen and spun four times, resulting in No, Yes, Yes, No.

- $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{1}{4}$: $\frac{9}{256}$
- $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{1}{2}$: $\frac{16}{256}$
- $P(\text{No, Yes, Yes, No})$ if $\pi = \frac{3}{4}$: $\frac{9}{256}$

Which spinner would you guess was spun? **Spinner B, because the probability of the observed data is the highest**

Likelihood

Likelihood: the probability of the observed data, regarded as a function of the parameters



$\pi = P(Yes) = \frac{1}{4}$ $\quad \pi = \frac{1}{2}$ $\quad \pi = \frac{3}{4}$

$\pi =$ **parameter value**	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$
$P(\text{No}, \text{No}) = \text{likelihood} = L(\pi)$	$\frac{9}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

Maximum likelihood

Maximum likelihood principle: estimate the parameters to be the values that maximize the likelihood

$\pi =$ **parameter value**	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$
$P(\text{No}, \text{No}) = \text{likelihood} = L(\pi)$	$\frac{9}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

Maximum likelihood estimate: $\widehat{\pi} = \frac{1}{4}$

Maximum likelihood

Maximum likelihood principle: estimate the parameters to be the values that maximize the likelihood

Steps for maximum likelihood estimation:

- *Likelihood:* For each value of the parameter, compute the probability of the observed data, $P(\mathrm{data})$
- *Maximize:* Find the parameter value that gives the largest $P(\mathrm{data})$

Maximum likelihood for logistic regression

Putting data: Putt at lengths 3, 4, 5, 6, and 7 feet. Record whether you make the put ($y = 1$) or miss ($y = 0$). Let $\pi = P(y = 1)$.

Regression model:

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \text{ length} \qquad \pi = \frac{\exp\{\beta_0 + \beta_1 \text{ length}\}}{1 + \exp\{\beta_0 + \beta_1 \text{ length}\}}$$

Observed data: (length = 3, make), (length = 3, make), (length = 4, make), (length = 5, miss), (length = 7, miss)

$P(\text{make} \mid \text{length}=3) = \frac{\exp\{3 - 0.5(3)\}}{1+\exp\{3-0.5(3)\}} = 0.818$

Maximum likelihood for logistic regression

Putting data: Putt at lengths 3, 4, 5, 6, and 7 feet. Record whether you make the put ($y = 1$) or miss ($y = 0$). Let $\pi = P(y = 1)$.

Regression model:

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \text{ length} \qquad \pi = \frac{\exp\{\beta_0 + \beta_1 \text{ length}\}}{1 + \exp\{\beta_0 + \beta_1 \text{ length}\}}$$

Observed data: (length = 3, make), (length = 3, make), (length = 4, make), (length = 5, miss), (length = 7, miss)

Suppose $\beta_0 = 3, \ \beta_1 = -0.5$. What is $P(\text{data})$ (aka $L(\beta_0, \beta_1)$)?

- need $P(\text{make} \mid \text{length}=3)$, $P(\text{make} \mid \text{length}=4)$, ... etc.

- $$\underbrace{P(\text{data})}_{L(\beta_0, \beta_1)} = P(\text{make} \mid \text{length}=3) \cdot P(\text{make} \mid \text{length}=3) \cdot P(\text{make} \mid \text{length}=4) \cdot (1 - P(\text{make} \mid \text{length } 5)) \cdot (1 - P(\text{make} \mid \text{length}=7))$$

Maximum likelihood for logistic regression

Regression model:

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \text{ length} \qquad \pi = \frac{\exp\{\beta_0 + \beta_1 \text{ length}\}}{1 + \exp\{\beta_0 + \beta_1 \text{ length}\}}$$

Observed data: (length = 3, make), (length = 3, make), (length = 4, make), (length = 5, miss), (length = 7, miss)

Suppose $\beta_0 = 3, \ \beta_1 = -0.5$. What is $P(\text{data})$ (aka $L(\beta_0, \beta_1)$)?

Length	**3**	**4**	**5**	**6**	**7**
π	$\frac{\exp\{3 - 0.5(3)\}}{1 + \exp\{3 - 0.5(3)\}} = 0.818$	0.731	0.622	0.5	0.378

$$\frac{\exp\{3 - 0.5(4)\}}{1 + \exp\{3 - 0.5(4)\}}$$

Maximum likelihood for logistic regression

Regression model:

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 \text{ length} \qquad \pi = \frac{\exp\{\beta_0 + \beta_1 \text{ length}\}}{1 + \exp\{\beta_0 + \beta_1 \text{ length}\}}$$

Observed data: (length = 3, make), (length = 3, make), (length = 4, make), (length = 5, miss), (length = 7, miss)

Suppose $\beta_0 = 3, \ \beta_1 = -0.5$. What is $P(\text{data})$ (aka $L(\beta_0, \beta_1)$)?

Length	3	4	5	6	7
π	$\frac{\exp\{3 - 0.5(3)\}}{1 + \exp\{3 - 0.5(3)\}} = 0.818$	0.731	0.622	0.5	0.378

$$L(\beta_0, \beta_1) = P(\text{data}) =$$

$$(0.818)(0.818)(0.731)(1 - 0.622)(1 - 0.378) = 0.115$$

Class activity

Work on the class activity (handout).

Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

What's the probability of the observed data, if $\beta_0 = 3$ and $\beta_1 = -0.5$?

Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

Length	3	4	5	6	7
π	$\frac{\exp\{3 - 0.5(3)\}}{1 + \exp\{3 - 0.5(3)\}} = 0.818$	0.731	0.622	0.5	0.378

$L(\beta_0, \beta_1) = P(\text{data}) =$

$$(0.818)^3(0.731)^2(0.622)^3(1 - 0.622)^2(0.5)(1 - 0.5) \cdot (0.378)(1 - 0.378)^2$$

$= 0.000368$

Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

What's the probability of the observed data, if $\beta_0 = 4$ and $\beta_1 = -0.75$?

Class activity

Length	3	4	5	6	7
Number of successes	3	2	3	1	1
Number of failures	0	0	2	1	2

Length	3	4	5	6	7
π	$\frac{\exp\{4 - 0.75(3)\}}{1 + \exp\{4 - 0.75(3)\}} = 0.852$	0.731	0.562	0.378	0.223

$$L(\beta_0, \beta_1) = P(\text{data}) =$$

$$(0.852)^3(0.731)^2(0.562)^3(1 - 0.562)^2(0.378)(1 - 0.378) \cdot (0.223)(1 - 0.223)^2$$

$$= 0.000356$$

Maximum likelihood for logistic regression

Likelihood:

- For estimates $\widehat{\beta}_0$ and $\widehat{\beta}_1$, $\widehat{\pi} = \dfrac{\exp\{\widehat{\beta}_0 + \widehat{\beta}_1 x\}}{1 + \exp\{\widehat{\beta}_0 + \widehat{\beta}_1 x\}}$
- $L(\widehat{\beta}_0, \widehat{\beta}_1) = P(\text{data})$

Maximize:

- Choose $\widehat{\beta}_0, \widehat{\beta}_1$ to maximize $L(\widehat{\beta}_0, \widehat{\beta}_1)$

> In the class activity examples, we have only considered a few different possibilities for $\widehat{\beta}_0$ and $\widehat{\beta}_1$. In reality, we consider all real numbers as possible values.

Fitting logistic regression

Logistic regression with maximum likelihood

- **Model:** $\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 x$
- **Measure of fit:** $L(\widehat{\beta}_0, \widehat{\beta}_1) =$ probability of observed data for estimates $\widehat{\beta}_0, \widehat{\beta}_1$
- **Optimize:** choose $\widehat{\beta}_0, \widehat{\beta}_1$ to maximize $L(\widehat{\beta}_0, \widehat{\beta}_1)$

Linear regression with least squares

- **Model:** $y = \beta_0 + \beta_1 x + \varepsilon$
- **Measure of fit:** $SSE = \sum_{i=1}^{n}(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i)^2$
- **Optimize:** choose $\widehat{\beta}_0, \widehat{\beta}_1$ to minimize SSE